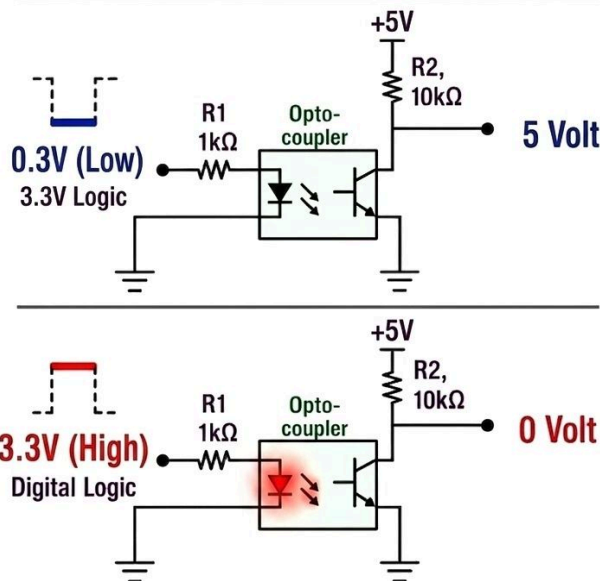




HIGH-SPEED DIGITAL ISOLATOR PRINCIPLE



Electronics Education

4 maj kl. 15:27 · 🌐

A digital isolator (optocoupler) transfers signals between two circuits without a direct electrical connection. On the left side, a 3.3V logic signal drives an internal LED through resistor R1. When the input is LOW (around 0.3V), the LED is OFF, so no light reaches the internal phototransistor. As a result, the transistor stays OFF, and the output is pulled up to 5V through R2. This gives a HIGH output while keeping both grounds electrically isolated.

When the input goes HIGH (3.3V), current flows through the LED, turning it ON. The emitted light activates the phototransistor on the output side, allowing current to flow to ground. This pulls the output down to 0V (LOW). In this way, the signal is transferred using light instead of a wire, providing noise immunity and protecting sensitive circuits from voltage spikes or ground differences.

#ElectronicsEducation #ElectronicsRD #Optocoupler #DigitalIsolation #EmbeddedSystems Visa mindre

👍 825 💬 13 ➦ 96

Mest relevant ▼



Författare

Electronics Education

Youtube: https://youtube.com/@electronicseducation?sub_confirmation=1... Visa mer




ElectronicsRD

YOUTUBE.COM
Electronics Education



2 v **Gilla** Svvara

Visa 1 svar

**Nicu Florica**

Usual leds in opyocoupler had thredhold for be on at 1.5V not 3V.. check 4N25 datasheet

2 v **Gilla** Svvara Redigerad

**Ramses Youhana**Just a note. When the NPN transistor is turned on and is in saturation mode, the Voltage across its Collector to Emitter is t... **Visa mer**

2 v **Gilla** Svvara Redigerad 

**Alfredo Torres**

6N137 har ingen transistor vid utgången, det är en And-krets, därför har den hög hastighet

2 v **Gilla** Svvara Visa ursprungsversion (Spanska)

**Dale Anthony**

Not that high speed without totem pole...

2 v **Gilla** Svvara 

**Roger Pry**

Iso-optolator

2 v **Gilla** Svvara



Skriv en kommentar ...

